

Oluwatobi Melvyn Mayungbo

Data Scientist, Machine Learning and Statistics

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SUMMARY

Data scientist with a B.Sc. in Statistics who takes models to production. Built an hourly demand forecasting system for city taxi zones with quantile forecasts, cost-optimal decisions and drift monitoring, an optimiser that picks the best legal squad under budget rules, and a fraud model served in milliseconds.

SKILLS

Payments and Risk: Demand Forecasting, Transaction Fraud Detection, Credit Risk Decisioning, Payment Reconciliation, Double-Entry Ledgers and Idempotency, Regulatory Compliance (CBN/NDPA), Web3 Fraud Risk Scoring

Statistics and Causal Inference: Time-Series Backtesting, Statistical Inference, Experiment Design, Power Analysis, CUPED, Causal Inference (S/T/X-Learners, Causal Forests), Uplift Modeling, Qini and AUUC, Logistic Regression, WOE Scorecards

Machine Learning: LightGBM, PyTorch, Feature Engineering, Scikit-learn, XGBoost, Deep Learning, Sequence Models (GRU, TCN, Transformers), Model Calibration (Isotonic Regression, Temperature Scaling), SHAP

Engineering and MLOps: Python, SQL, Prefect, dbt-core, DuckDB, MLflow, Evidently, FastAPI, Docker, PostgreSQL, Redis, GCP Cloud Run, AWS ECS Fargate, Railway, CI/CD (GitHub Actions), Git, Pytest, Streamlit, React

AI and LLM Systems: Retrieval-Augmented Generation (RAG), Hybrid Search (BM25, Dense Embeddings, RRF), LLM Evaluation and Guardrails, Prompt Engineering, Citation Verification, Model Context Protocol (MCP), Hugging Face, ONNX Runtime

PROJECTS

Self-Operating Demand Forecasting Platform | [GitHub](#)

Tech: Python, Prefect, dbt-core, DuckDB, LightGBM, MLflow, Evidently, FastAPI, Docker, GitHub Actions

- Built an end-to-end hourly demand forecasting system for NYC taxi zones, producing 24-hour quantile forecasts through an orchestrated Prefect pipeline spanning ingestion, dbt/DuckDB transformations, backtesting, drift monitoring and conditional retraining.
- Modelled asymmetric business cost by penalising unmet demand at 3× idle supply and shipping the cost-optimal quantile rather than the median, cutting expected cost by 25% on an identical model and beating a seasonal-naive baseline by 32%.
- Validated with rolling-origin backtesting across 6 expanding-window folds against seasonal-naive and historical-mean baselines, reporting MASE and pinball loss, with automated leakage tests that fail the build if any feature reads past the forecast origin.

FPL AI Manager | [GitHub](#)

Tech: Python, SQLite, FastAPI, React, Docker, Render

- Built two models that play Fantasy Premier League against the live FPL API: a Manager that carries one squad all season under real transfer, budget and chip rules, and a Best XI optimiser that rebuilds the best legal squad each gameweek.
- Wrote a rules engine, an expected-points model over a multi-gameweek horizon and a leakage-free backfill; cached, rate-limited API client, scheduled jobs and one-command deployment, with 574 tests.

Sequence-Based Transaction Fraud Detection System | [GitHub](#)

Tech: Python, PyTorch, LightGBM, ONNX Runtime, FastAPI, Docker, SHAP

- Deployed real-time scoring by exporting to ONNX Runtime (2.9× faster than eager PyTorch at batch size 1) and serving via FastAPI, reaching 5.1 ms p99 end-to-end latency against a 50 ms budget, with append-only per-decision audit logging and attention-based attribution of the prior transactions driving each flag.
- Built and compared three deep sequence architectures (GRU, Temporal Convolutional Network and a causally masked Transformer encoder) over customer transaction histories, with learned embeddings for high-cardinality merchant (~2.9k vocabulary), MCC and device-type categoricals; all models under 115k parameters and CPU-trainable in under 6 minutes.

EXPERIENCE

Machine Learning Instructor, SQI College of ICT

March 2026 to Present

- Teach machine learning and deep learning to student cohorts, covering statistical foundations, classical ML algorithms, neural architectures, sequence models and production evaluation metrics.
- Instruct students on data analysis, feature engineering and model building with PyTorch, Scikit-Learn, Pandas and NumPy.

Internal Auditor, NISER

May 2024 to July 2024

- Analysed financial and operational records to identify inconsistencies and irregularities, performing manual anomaly detection across complex transactional datasets.

EDUCATION

B.Sc. Statistics, University of Ibadan

January 2021 to March 2025

Professional Diploma in Artificial Intelligence, SQI College of ICT

July 2025 to Present

CERTIFICATIONS

- Google Data Analytics Professional Certificate
- Google Advanced Data Analytics Professional Certificate
- Claude Code In Action